

BT305 Project – molconvert

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Objective:

The primary objective of this project is to develop a **robust CLI-based molecular structure conversion tool** that enables seamless transformation between different molecular file formats—specifically **PDB, SDF, and Z-matrix (ZMAT)**—using a unified internal representation.

At its core, the project aims to:

- Establish **internal coordinates (bond lengths, bond angles, dihedral angles)** as the central representation of molecular structures
- Ensure **lossless and consistent conversion** between multiple formats through a **JSON-based intermediate representation (MoleculeIC)**
- Provide a **reliable reconstruction mechanism** (via the NeRF algorithm) that preserves structural accuracy with extremely low RMSD ($< 0.0001 \text{ \AA}$)
- Enable **analysis and validation of molecular structures**, including round-trip accuracy checks

In essence, the tool is designed not just for format conversion, but for **structural understanding, validation, and computational analysis** of molecular data.

Comparison with Open Babel & Unique Selling Points (USP):

This tool is conceptually similar to Open Babel, which is widely used for molecular file format interconversion. Both tools allow users to convert between formats like PDB and SDF.

However, **molconvert** differentiates itself through several strong USPs:

Key USP 1: Built-in RMSD Calculation via CLI

A major advantage is the ability to compute RMSD directly within the CLI:

```
rmsd structure1.pdb structure2.pdb
```

```
rmsd input.pdb --self
```

- Supports **round-trip validation (PDB → IC → PDB)**
- Allows **filtered RMSD calculations** (e.g., backbone atoms only)
- Provides **per-atom deviation analysis**
- uses **Kabsch algorithm**, which ensures optimal alignment of molecular structures by minimizing the least-squares deviation between corresponding atoms. This makes the RMSD invariant to rotation and translation.

In contrast, Open Babel does not provide such **direct, built-in RMSD analysis in a clean CLI workflow**.

Key USP 2: Internal Coordinate–Centric Architecture

Unlike Open Babel, which primarily operates on Cartesian coordinates and format-specific logic, this tool:

- Uses a **single, unified internal representation (MoleculeIC)**
- Represents molecules in terms of **bond lengths, angles, and dihedrals**
- Ensures all conversions pass through this **consistent and interpretable intermediate layer**

Internal coordinates provide a representation that is invariant to global rotation and translation, making them particularly suitable for structural comparison and machine learning applications.

This makes the system more **modular, extensible, and mathematically grounded**.

Key USP 3: Internal Coordinate (IC) Statistics

The tool can compute statistical insights about molecular geometry:

convert input.pdb --summary

Outputs include:

- Mean and standard deviation of **bond lengths**
- Distribution of **bond angles**
- Range and spread of **dihedral angles**

This transforms the tool from a converter into an **analytical instrument**, which Open Babel does not emphasize.

Key USP 4: High-Fidelity Reconstruction (NeRF Algorithm)

- Uses the **NeRF (Natural Extension Reference Frame)** algorithm for reconstruction
- Achieves **extremely low RMSD (< 0.0001 Å)** in round-trip conversions

This guarantees **near-lossless structural recovery**, which is not always a focus in traditional converters.

Key USP 5: JSON-Based Transparent Internal Representation

- Internal structure stored as **human-readable JSON**

- Easy to integrate with **machine learning pipelines**
- Enables **inspection, debugging, and customization**

Open Babel does not expose such a **clean, developer-friendly internal representation**.

Key USP 6: Modular and Extensible Design

- Adding a new format requires only:
 - A parser (input)
 - A builder (output)
- No need to modify the core system

This makes the tool highly **scalable and future-proof**.

Conclusion

While this project shares its foundational goal with Open Babel—molecular format conversion—it goes significantly further by introducing:

- A **mathematically rigorous IC-based pipeline**
- **Integrated structural validation tools (RMSD)**
- **Analytical capabilities (IC statistics)**
- **High reconstruction accuracy**
- A **clean, ML-friendly architecture**

Overall, molconvert is not just a converter, but a **computational framework for molecular structure representation, analysis, and validation**.
